K-means with learned metrics

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Outline

- K-mean
- Metric learning
- K-mean with metric learning : Examples
- K-mean with metric learning : consistency results
- Applications

K-mean

Goal : Cluster data

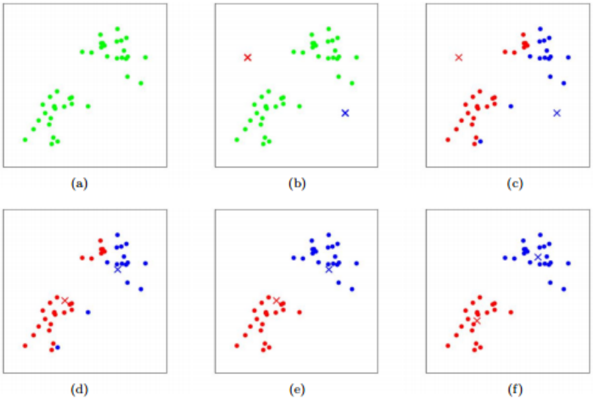

Figure – by Chris Piech.

K-mean

- population K-mean

$$\underset{\#A \leq k}{\operatorname{argmin}} \mathbb{E}\left[\min_{a \in A} ||X - a||^2\right]$$

K-mean

- population K-mean

$$\operatorname*{argmin}_{\#A\leq k} \mathbb{E}\left[\min_{a\in A} ||X-a||^2\right]$$

- empirical K-mean

$$\operatorname*{argmin}_{\#A\leq k} \frac{1}{n}\sum_{i=1}^{n} \min_{a\in A} ||X_i-a||^2$$

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Lloyd's Algorithm (1957)

1. Initialize *cluster centroids* $m_1, \cdots, m_k$
2. Repeat :
   - cluster $C_i =$ Voronoï cell of m_i, $i = 1, \cdots, k$
   - cluster centroids $m_i \leftarrow \text{mean}(C_i)$, $i = 1, \cdots, k$

K-mean theoretical guarantees

- Pollard, 1981 : $\exists!$ pop. K-mean $\Rightarrow$ strong consistency
- Lember, 2002 : treat the case of non uniqueness

Fréchet K-mean : extension to metric spaces

Let (M, d) be a metric space, and $X \in M$ random.

- population K-mean

$$\underset{\#A \leq k}{\operatorname{argmin}} \, \mathbb{E}\left[\min_{a \in A} d(X, a)^2\right]$$

- empirical K-mean

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Fréchet K-mean : extension to metric spaces

Let (M, d) be a metric space, and $X \in M$ random.

Also for $p \geq 1$.

- population K-mean

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Fréchet K-mean theoretical guarantees

- Lember, 2002.
- Jaffe, 2025 : adaptative for k (elbow method).

Link with the quantization of measure

An kth-order quantization q_k of a probability measure μ :

$$q_k \in \arg\min_{q \in \mathcal{P}_k} W_2(\mu, q),$$

where

$$\mathcal{P}_n = \left\{ \sum_i \alpha_i \delta_{x_i} \mid (x_1, \cdots, x_k) \in \mathbb{X}^k,\ \sum_{i=1}^k \alpha_i = 1 \right\}$$

is the set of probability supported on at most k points.

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is the set of probability supported on at most k points.

- this problem is equivalent to K-mean,
- q_k is supported on the set of k-centroids,
- in quantization, interested in $k >> 1$ to get $q_k \approx \mu$.

Metric learning

Metric learning

$$X_1, \cdots, X_n \in \mathbb{R}^D$$

Manifold hypothesis

$$X_1, \cdots, X_n \in \mathcal{M}^d \subset \mathbb{R}^D,$$
$$d << D.$$

Metric learning

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Philosophy

Instead of *measurable* maps from *probability spaces*

$$X : (\Omega, \mathbb{P}) \to (\mathbb{R}^D, \mu),$$

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$$X : (\Omega, dist, \mathbb{P}) \to (\mathcal{M}, d_{\mathcal{M}}, \mu).$$

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Goal : Learn the intrinsic metric

Metric learning

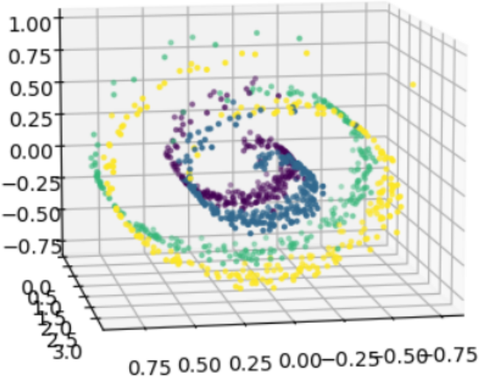


Figure – Swiss roll

Metric learning : Fermat / ISOMAP

$$\mathbb{X}_n = \{x_1, \cdots, x_n\} \subset \mathbb{R}^D, \quad \alpha \geq 1$$

For $x, y \in \mathbb{X}_n$, $r_{xy} = (x_1, \cdots, x_l)$ a $\mathbb{X}_n$-path from x to y,

$$length(r_{xy}) = \sum_{i=1}^{l-1} ||x_{i+1} - x_i||^\alpha$$

$$d_{\mathbb{X}_n,\alpha}(x, y) = \inf_{r_{xy}} length(r_{xy})$$

- $\alpha = 1$, neighborhood graph : ISOMAP
- $\alpha > 1$, complete graph : Fermat distance

Metric learning : Fermat distance

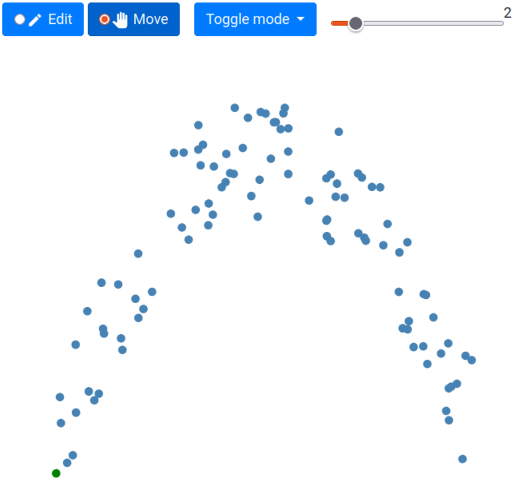
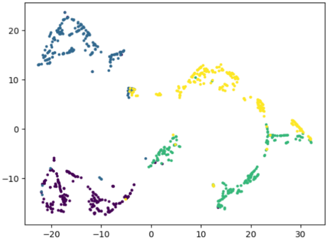
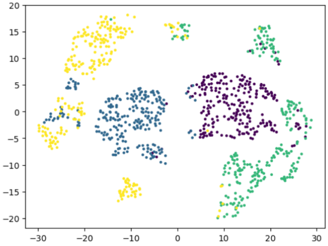


Figure – https://www.aristas.com.ar/fermat/index.html

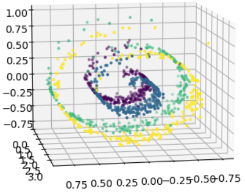
Metric learning : Fermat distance



(a) tSNE - Fermat ($\alpha = 3$)



(b) tSNE - Euclidien



Metric learning : Fermat distance

(Hwang-Damelin-Hero, 16 ; Groisman-Jonckheere-Sapienza 18)

Let $X_1, \cdots, X_n \in \mathcal{M}^d \subset \mathbb{R}^D$, i.i.d with density $f > 0$.
If $\mathcal{M}$ is smooth enough, then $\exists c_\alpha > 0$ such that almost surely,

$$n^{(\alpha-1)/d} d_{\mathbb{X}_n,\alpha} \underset{n\to\infty}{\longrightarrow} c_\alpha \, d_{f,\alpha}$$

where

$$d_{f,\alpha}(x,y) = \inf_\gamma \int_\gamma f^{(1-\alpha)/d} d\ell, \quad x, y \in \mathcal{M}.$$

is the population Fermat distance.

Metric learning : Fermat distance

On the uniqueness of the Frechet mean in Fermat distance

Frechet mean in Fermat distance

- X in $(\mathbb{R}^\ell, |\cdot|)$ with density $f > 0$

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Theorem (Groisman, Jonckheere, S, Sued, 2026)

If f is log-concave, then $(\mathbb{R}^\ell,\ f^{2(1-\alpha)/d}\, I)$ is Hadamard.

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Corollary

If f is log-concave, then X admits *at most* one Fréchet mean w.r.t. the Fermat distance.

K-mean with metric learning : Example

Kmean with Fermat

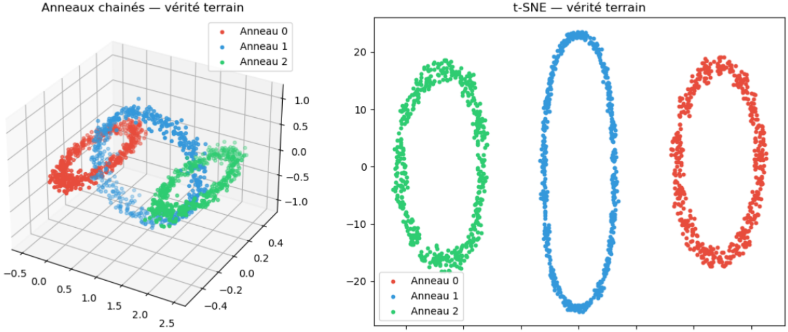


Figure – linked rings

Kmean with Fermat

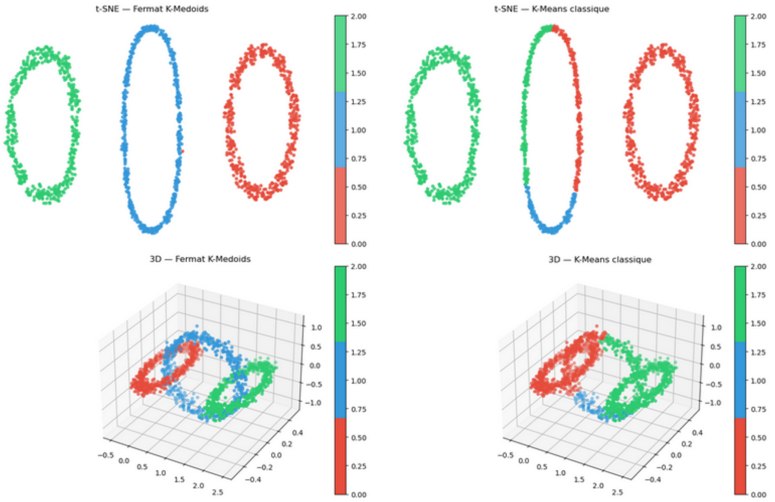
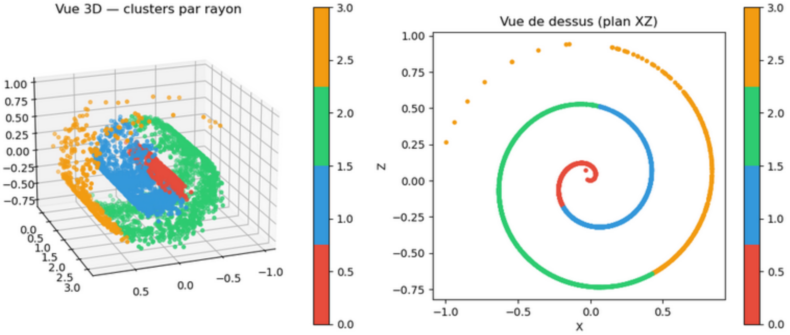


Figure – accuracy : 0.99 vs 0.75

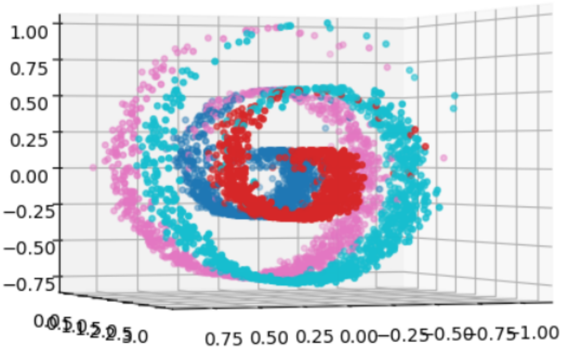
Kmean with Fermat



	ARI	Accuracy
Fermat K-Medoids	0.609	0.741
K-Means classique	0.083	0.381

Figure – alpha = 6, k = 7

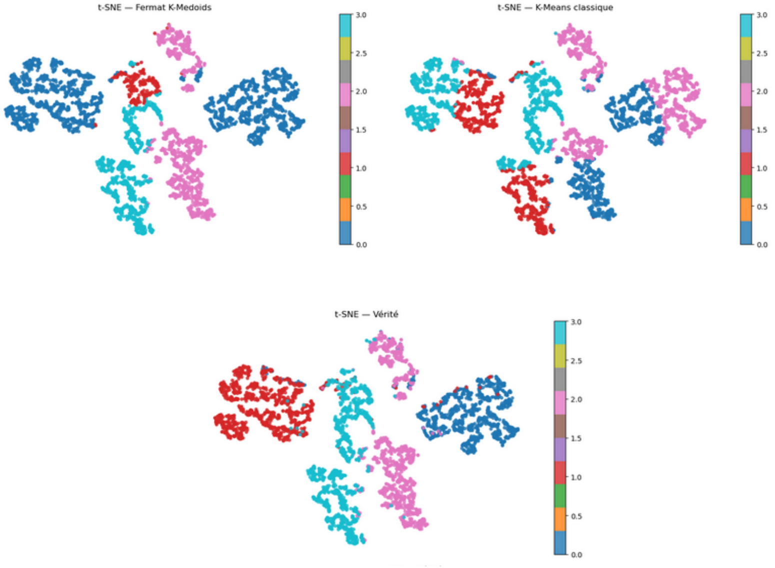
Kmean with Fermat



	ARI	Accuracy
Fermat K-Medoids	0.603	0.682
K-Means classique	0.302	0.492

Figure – alpha = 4, k = 10

Kmean with Fermat



K-mean with metric learning : Consistency results

K-mean with metric learning

Goal

Establish consistency of K-mean with empirical distance, in the most general setting.

- population K-mean

$$\operatorname*{argmin}_{\#A \leq k} \mathbb{E}\left[\min_{a \in A} d(X, a)^2\right]$$

- empirical K-mean with empirical distance

$$\operatorname*{argmin}_{\#A \leq k} \frac{1}{n} \sum_{i=1}^{n} \min_{a \in A} d_n(X_i, a)^2$$

Hausdorff distance

Def : Hausdorff distance

For $A, B \subset (\mathbb{X}, d)$,

$$d_H(A, B) = \max \left\{ \max_{a \in A} d(a, B), \max_{b \in B} d(b, A) \right\}.$$

It is the smallest ε such that $A \subset B^\varepsilon$ and $B \subset A^\varepsilon$.

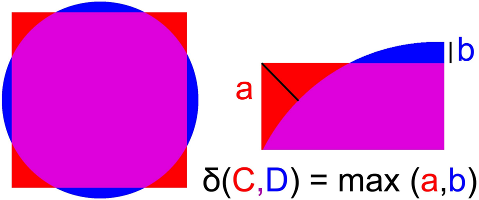


Figure – Wikipedia

mGH convergence

Def : ε-isometry

A map $h : (\mathbb{X}, d_{\mathbb{X}}) \to (\mathbb{Y}, d_{\mathbb{Y}})$ between compact metric spaces

- $\forall x, x' \in \mathbb{X}$ we have $|d_{\mathbb{X}}(x, x') - d_{\mathbb{Y}}(h(x), h(x'))| < \varepsilon$
- $\forall y \in \mathbb{Y}$ we have $d(y, h(\mathbb{X})) < \varepsilon$.

Def : mGH convergence

We say that $\mathcal{X}_n \overset{mGH}{\to} \mathcal{X}$ iff

- there exist numbers $\varepsilon_n \to 0$
- there exist ε_n-isometries $h_n : \mathbb{X}_n \to \mathbb{X}$
- $(h_n)_{\#}(\mu_n) \to \mu$ weakly on $\mathbb{X}$ when $n \to \infty$

mGH convergence

Def : mGH convergence

We say that $(\mathbb{X}_n, d_n, \mu_n) \overset{mGH}{\rightarrow} (\mathbb{X}, d, \mu)$ iff

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- $(h_n)_{\#}(\mu_n) \rightarrow \mu$ weakly on $\mathbb{X}$ when $n \rightarrow \infty$

If $(\mathbb{X}_n, d_n) \subset (\mathbb{X}, d)$, then mGH convergence equivalent to $d_H(\mathbb{X}_n, \mathbb{X}) \rightarrow 0$ and $\mu_n \rightharpoonup \mu$.

K-mean with metric learning

Theorem (Groisman, Jonckheere, S, Sued, 2026)

If $(\mathbb{X}_n, d_n, \mu_n)$ converges in measured Gromov-Hausdorff topology to $(\mathbb{X}, d_{\mathbb{X}}, \mu)$, then

$$\lim_{n\to\infty} \max_{S_n \in \mathcal{S}_{\mathcal{X}_n}(k)} \min_{S \in \mathcal{S}_{\mathcal{X}}(k)} d_H(h_n(S_n), S) = 0.$$

- $\mathcal{S}_{\mathcal{X}_n}(k)$ is the set of all sets of k-mean centroids for $\mathbb{X}_n$.
- $\mathcal{S}_{\mathcal{X}}(k)$ is the set of all sets of k-mean centroids for $\mathbb{X}$.
- d_H is the Hausdorff distance between subsets of $(\mathbb{X}, d)$.
- $h^n \colon \mathbb{X}_n \to \mathbb{X}$ is a sequence of ε_n-isometries, with $\varepsilon_n \to 0$.

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When $\mathbb{X}_n \subset \mathbb{X}$, h_n is the identity.

What does it mean ? ?

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$\forall \varepsilon > 0$, there exists $N \in \mathbb{N}$ such that for all $n \geq N$,

for every set of k-barycenters $S_n \in \mathcal{S}_{\mathcal{X}_n}(k)$ of $(\mathbb{X}_n, d_n, \mu_n)$,

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No-false-positive guarantee !

K-mean with metric learning

Idea of proof

Show that all subsequences of $(S_n)_n$ with $S_n \in \mathcal{S}_{\mathcal{X}_n}(k)$ have a subsequence converging towards an element of $\mathcal{S}_{\mathcal{X}}(k)$.

Technically, it is based on a result by Jaffe (2025), establishing the continuity of

$$\Phi_{\mathbb{X}} : C_k(\mathbb{X}) \times \mathcal{P}(\mathbb{X}) \to \mathbb{R}$$
$$\Phi_{\mathbb{X}}(S, \mu) := \int_{\mathbb{X}} \min_{x \in S} d_{\mathbb{X}}^2(y, x)\, \mu(dy)$$

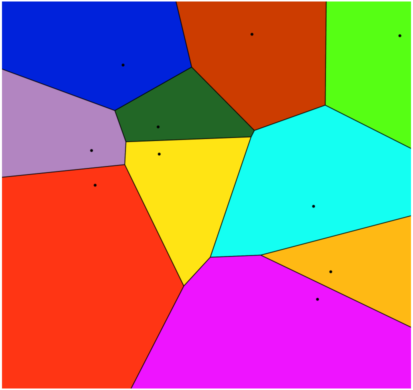
where $C_k(\mathbb{X}) = \{A \subset \mathbb{X} \,|\, \#A \leq k\}$.

Consistency of clusters

Consistency of clusters

- $S = \{b_1, \cdots, b_k\}$ the k centroids
- $\mathcal{V}(S) = \{V_{b_1}, \cdots, V_{b_k}\}$ the Voronoi cells (clusters)

$$V_{b_i} := \{x \in \mathbb{X} \mid \forall b \in S,\ d(x, b_i) \leq d(x, b)\}$$



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Theorem (Groisman, Jonckheere, S, Sued, 2026)

Assume that

$$(\mathbb{X}_n, d_n, \mu_n) \stackrel{mGH}{\to} (\mathbb{X}, d, \mu).$$

Then

$$\forall \varepsilon > 0,\ \exists N \text{ such that } \forall n \geq N,$$

$$\forall V \in \mathcal{V}(S_n),\ \exists W \in \mathcal{V}(S) \quad \text{with} \quad V \subset W^{\varepsilon}.$$

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Again a no-false-positive guarantee !

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- We show that the cell of every $x_n \in S_n$ is contained in the enlarged Voronoi cell of some 'true' centroid y for some $\delta > 0$:

$$W(y, \delta) := \{x \in \mathbb{X} \,|\, \forall b' \in S,\ d(x, b) \leq d(x, b') + \delta\}\,.$$

Consistency of clusters

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$$W(y, \delta) := \{x \in \mathbb{X} \,|\, \forall b' \in S,\ d(x, b) \leq d(x, b') + \delta\}\,.$$

- We show continuity of the enlarged Voronoi cells in Hausdorff distance :

$$d_H(W(y), W(y, \delta)) \to 0, \quad \text{when} \quad \delta \to 0.$$

where $W(y) = W(y, 0)$ is the true Voronoi cell.

When does mGH convergence hold in practice ?

mGH in practice ?

Proposition

Let $\mathbb{X}_n := \{X_1, \cdots, X_n\}$ i.i.d from μ on $(\mathbb{X}, d)$ compact. Let μ_n be the empirical measure on $\mathbb{X}_n$. If

$$\sup_{x,x' \in \mathbb{X}_n} |d_n(x,x') - d(x,x')| \to 0,$$

and

$$\sup_{x \in \mathbb{X}} d(x, \mathbb{X}_n) \to 0,$$

then $(\mathbb{X}_n, d_n, \mu_n)$ converge almost surely to $(\mathbb{X}, d, \mu)$ in the measured Gromov-Hausdorff topology

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Remark

The second condition is satisfied when μ is fully supported on a Riemannian manifold and absolutely continuous w.r.t. the volume measure.

Application : Wasserstein K-mean

Application : Wasserstein

- $\mu_1, \cdots, \mu_{10}$ Gaussian measures on $\mathcal{M}$ the Swiss roll,
- 4 clusters,
- $X_1^i, \cdots, X_{100}^i \sim \mu_i$,
- $\mathbb{X}_n := \{X_j^i\} \subset \mathbb{R}^3$,
- d_n learned with ISOMAP on $\mathbb{X}_n$ + a uniform sample on $\mathcal{M}$ (to connect the graph),
- $d_W^{(n)}$ Wasserstein distance on $(\mathbb{X}_n, d_n)$,
- ν_n uniform distribution on $\{\hat{\mu}_1, \cdots, \hat{\mu}_{10}\}$

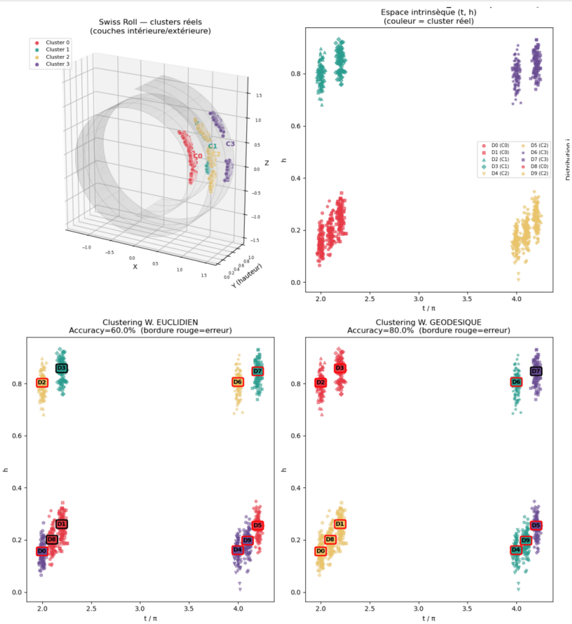
Application : Wasserstein

- $\mu_1, \cdots, \mu_{10}$ Gaussian measures on $\mathcal{M}$ the Swiss roll,
- 4 clusters,
- $X_1^i, \cdots, X_{100}^i \sim \mu_i$,
- $\mathbb{X}_n := \{X_j^i\} \subset \mathbb{R}^3$,
- d_n learned with ISOMAP on $\mathbb{X}_n$ + a uniform sample on $\mathcal{M}$ (to connect the graph),
- $d_W^{(n)}$ Wasserstein distance on $(\mathbb{X}_n, d_n)$,
- ν_n uniform distribution on $\{\hat{\mu}_1, \cdots, \hat{\mu}_{10}\}$

Our result

K-mean on $(\mathcal{P}(\mathbb{X}_n), d_n, \nu_n)$ is consistent.

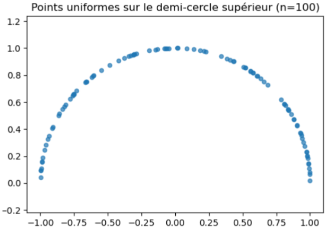
Application : Wasserstein



Application : Computation of intrinsic Fréchet barycenter

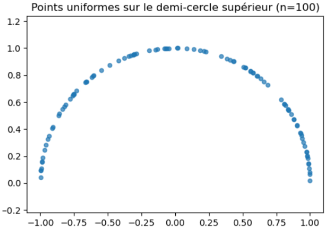
Intrinsic Fréchet barycenter

- $\mathbb{X}_n := \{X_1, \cdots, X_{100}\}$ uniform on the semi-circle
- d_n learned with ISOMAP
- ν_n uniform on $\mathbb{X}_n$



Intrinsic Fréchet barycenter

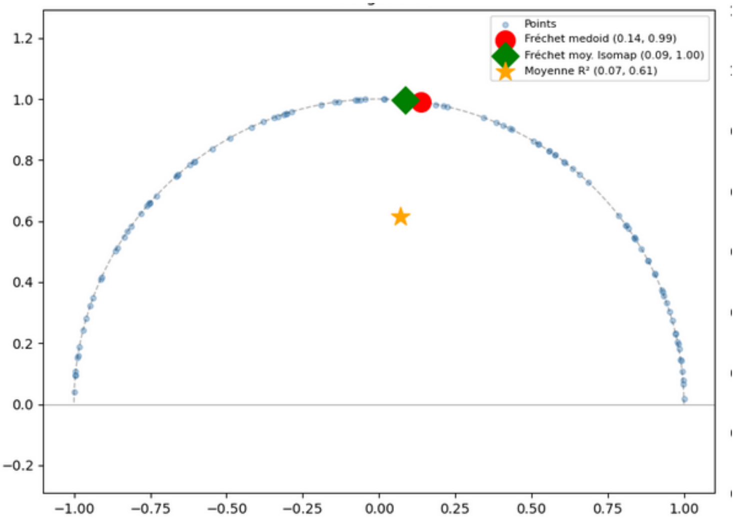
- $\mathbb{X}_n := \{X_1, \cdots, X_{100}\}$ uniform on the semi-circle
- d_n learned with ISOMAP
- ν_n uniform on $\mathbb{X}_n$



Our result

Fréchet mean on $(\mathbb{X}_n, d_n, \nu_n)$ is strongly consistent (converges to $(\frac{\pi}{2}, 0)$).

Intrinsic Fréchet barycenter



Thanks for listening !